

Experimental study of the performance of liquid cooling tank used for single-phase immersion cooling data center

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ABSTRACT

High efficiency cooling method in data center (DC), such as single-phase immersion cooling system, has received more attention. For practical application, server servers would be placed in a liquid cooling tank where the coolant should be evenly transferred to every server. Therefore, the performance of liquid cooling tank is crucial to the operation of DC, especially for the temperature uniformity among GPUs. In this study, the liquid cooling tank was designed and the impact of inlet temperature, coolant flowrate, and the load ratio of server on the performance was experimentally conducted. Results showed that, decreasing inlet temperature of coolant and the load ratio of GPUs, and increasing the flowrate are benefit to decrease the average GPU temperature. The decrease of dynamic viscosity and the increase of thermal conductivity is also helpful to the heat exchange. When the inlet temperature increased from 20 °C to 50 °C, the outlet temperature was increased by 30 °C; however, the average GPU temperature was only increased by 21 °C. The designed liquid cooling tank showed a good temperature uniformity, where the temperature deviation was within 5 °C and the square deviation was in the range of 1.0–1.5.

1. Introduction

The swift evolution of technologies like big data and cloud computing has spurred a noticeable migration towards high-density and integrated data centers (DCs). However, this trend exacerbates energy challenges [1,2]. Globally, DCs emerge as formidable entities in the realm of energy consumption [3]. According to IEA report, it consumed the global electricity consumption by 1.15 % at 2021 [4]. Notably, the energy consumed by cooling systems constitutes a substantial portion, ranging from 30 to 50 % of DCs' total electricity usage for air-based cooling data center [5]. Therefore, enhancing the cooling efficiency for DCs grows increasingly imperative.

Regardless of the type of cooling system, maintaining the temperature uniformity for high heat flux components, such as CPUs and GPUs, is vitally important. A lack of temperature uniformity can lead to the occurrence of hot spots, resulting in server shutdowns, or even worse, server burnouts [6]. Many efforts have been made to improve temperature uniformity, which, however, is conducted based on the air-based cooling data centers currently [7]. The data center with bad temperature uniformity primarily stems from three

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Table 1

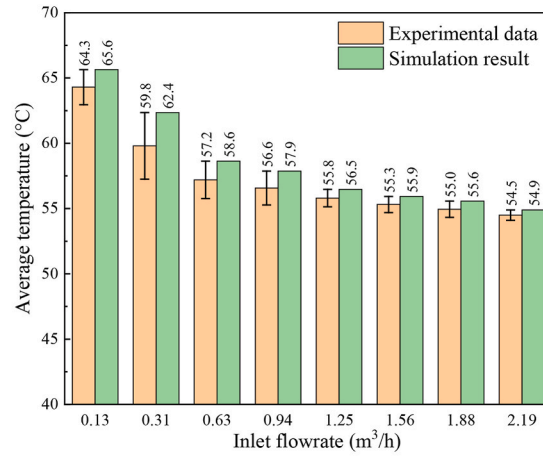
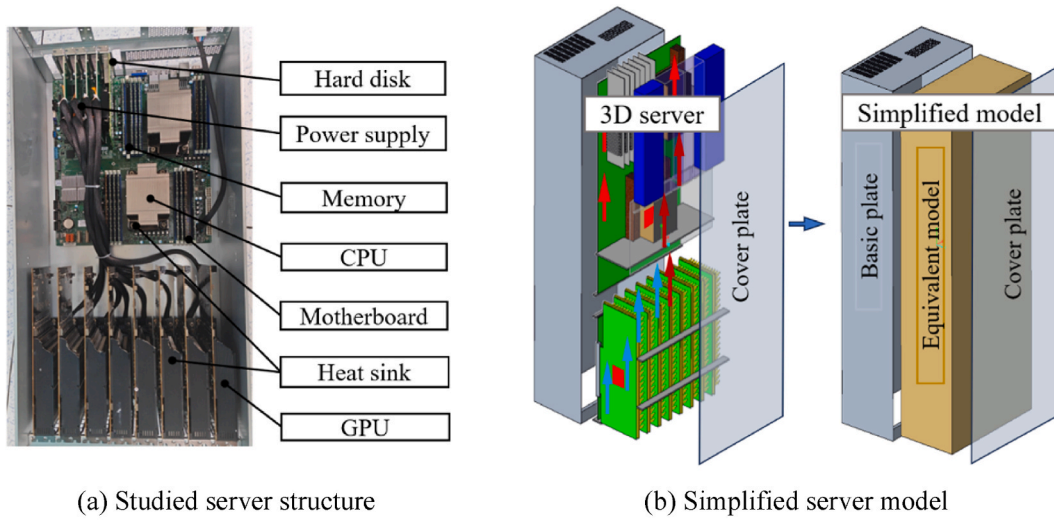
Current studies for the performance of single-phase immersion cooling data center.

Research content	Coolant	Heat load	Performance indicators	Parameters/variables	Remarks	Reference
Coolant	S5X; SmartCoolant; ThermoSafe R	Server: 620 W	Average temperature	Flowrate: 0.35–0.55 LPM; Inlet temperature of coolant: 30–45 °C.	The dynamic viscosity of the coolant increased from 4.6 to 9.8 mPa s, and the average temperature of the IT equipment increased by 6 %.	Hnayno et al. [18]
Coolant	Three different coolants	CPU: 270W × 2; GPU: 170W × 2; Server: 1421.5 W	Nusser number; Pressure drop	Flowrate: 5.1 L/s; Inlet temperature of coolant: 20 °C	The average Nusser number and pressure drop of single-phase immersion liquid cooling tank in electronic fluorinated liquids with low dynamic viscosity increased by 57.3 % and decreased by 59 %, respectively.	Chen et al. [2]
Coolant	Mineral oil; Synthetic oil; Silicone oil	CPU: 200W × 2; Server: 860 W	Nusser number; Pressure drop	Flowrate: 67.36 g/s; Inlet temperature of coolant: 30 °C.	Compared to mineral oil and synthetic oil, the single-phase immersion cooling system using silicone oil with higher dynamic viscosity exhibits the largest reduction of 24.4 % in average Nusselt number, while experiencing a significant increase of 7.4 times in flow resistance losses.	Wang et al. [23]
Coolant	Mineral oil based SiC nanofluids	CPU: 60W; Server: 60W	Convective heat transfer coefficient	Flow velocity: 0.3 m/s; Inlet temperature of coolant: 25 °C.	Under the same working conditions, the convective heat transfer coefficient of SiC nanofluid is increased by about 30 % compared with that of pure mineral oil	Luo et al. [24]
Performance optimization – server level	Electronic fluorinated liquids	CPU: 270W × 2; Server: 1331.9 W	PUE; Coolant average temperature	Flowrate: 3.7–67.82 LPM Inlet temperature of coolant: 20–40 °C.	Compared to buoyancy-driven SPIC systems, the average temperature of immersion coolants and PUE in pump-driven single-phase immersion cooling systems reduces by 55.5 and 11.6 %, respectively.	Huang et al. [25]
Performance optimization – server level	Mineral oil	CPU: 95W × 2; Server: 190 W	pPUE; Thermal resistance	Flowrate: 0.5–2.5 LPM Inlet temperature of coolant: 30–50 °C.	Comparing with results from baseline tests performed with traditional air cooling, the technology shows a 34.4 % reduction in the thermal resistance of the system. Overall, the cooling loop was able to achieve partial power usage effectiveness (pPUECooling) values as low as 1.03.	Eiland et al. [26]
Performance optimization – server level	Novec 7100	CPU: 95W; Server: 95W	CPU temperature	Flow velocity: 0.4–1.2 m/s.	1. Aluminum and copper heat sinks are compatible with the working fluid 2. A higher flowrate caused a more uniform temperature distribution.	Cheng et al. [16]
Performance optimization – server level	FC-40	CPU: 200–600W; Server: 200–600W	Thermal resistance; Temperature	Flowrate: 1–3 LPM; Inlet temperature of coolant: 15–35 °C	The thermal resistance and temperature of the T configuration are reduced by 12.6 % and 0.5–2.8 °C, respectively, compared to the Z configuration.	Muneeshwaran et al. [20]
Performance optimization – server level	EC-110	CPU: 120W × 2; Server: 270W	CPU maximum temperature; Power	Flowrate: 0.1–0.5 m ³ /h; Inlet temperature of coolant: 34 °C	After optimization, the maximum temperature rise of the server is reduced from 45.6 °C to 35.2 °C, and the power consumption is also reduced from 1.2 kW to 0.9 kW.	Li et al. [21]
Performance optimization – server level	Noah@3000 A, PAO-4	CPU: 300W; Server: 300W	Thermal resistances	Flowrate: 1–3 LPM; Inlet temperature of coolant: 35 °C	Immersing the cold plate in liquid enhances convective heat transfer, reducing thermal resistances by 7.2 % and 9.4 % for Noah@3000A and PAO-4 at low flow rates.	Zhang et al. [27]
Performance optimization – server level	FC-40 PAO-6	CPU: 200–600W; Server: 200–600W	Temperature of server; Thermal resistances	Flowrate: 1–3 LPM; Inlet temperature of coolant: 15–35 °C	The thermal resistance and temperature of the server will rise monotonically with the increase of the by-pass fluid. Without bypass, the cooling performance of the system is improved.	Shrigondekar et al. [28]
Performance optimization	HFE-7100	CPU: 20–40W/cm ²	Temperature difference of chip	Flowrate: 0.3–2.8 LPM	The embedded gradient distributed microneedle fin array is designed and fabricated, which can reduce the surface temperature difference by about 30 %.	Feng et al. [29]

(continued on next page)

Table 1 (continued)

Research content	Coolant	Heat load	Performance indicators	Parameters/variables	Remarks	Reference
– server level				Inlet temperature of coolant: 20 °C		
Performance optimization –server level	EC-100	CPU: 95W × 2 Server: 500W	CPU temperature	Flowrate: 0.5–1 GPM;	Using an oil inlet temperature of 45 °C and a volume flow rate of 1GPM keeps the CPU temperature around 70 °C and enables a more even distribution of the temperature field inside the server.	Gandhi et al. [30]
Performance optimization -Server level	Opticool 872,552	CPU: 100–110W Server: 100–110W	Average temperature of the server	Inlet temperature of coolant: 15–25 °C Inlet temperature of coolant: 30–60 °C; Environmental temperature: 25–55 °C	The average temperature of the server increased drastically with the environmental temperature however, the power consumption of the server remains unchanged.	Bansode et al. [31]
Performance optimization -Tank level	Electronic fluorinated liquids	CPU 350W × 2; Server: 850W × 2	Chip temperature; Thermal resistance; PUE	Flowrate: 2–8 LPM; Inlet temperature of coolant: 18 °C	The anti-gravity flow scheme reduced the chip housing temperature and thermal resistance by 33.8 % and 55.6 %, respectively, while reducing the PUE by 1.4 %.	Huang et al. [32]
Performance optimization -Tank level	Silicone oil 20; Silicone oil 50; Soybean oil; FC3283; FC43	CPU: 140W × 2 Server: 440W × 32	PUE	Flow velocity: 0.005–1 m/s; Inlet temperature of coolant: 15–35 °C	This cooling technology reduces the system PUE below 1.04 and improves space utilization and system stability.	Matsuoka et al. [33]



(c) Experimental results for single server and model validation for the simplified CFD model

Fig. 1. Studied server structure and the simplified process.

aspects: non-uniform airflow distribution, unbalanced airflow supply and demand, and airflow loss [8]. Non-uniform airflow distribution can be addressed by modifying the layout of the Computer Room Air Conditioners (CRACs), and optimizing the height of the plenum. For instance, VanGilder et al. [9] compared different CRAC layouts and found that placing two CRAC units opposite each other yielded the best performance. Beitelmal et al. [10] determined that maintaining the height of the underfloor plenum within the range of 0.76 m–0.91 m resulted in superior uniformity and minimized hotspot temperatures. The issue of unbalanced airflow supply and demand can be mitigated by increasing the total volume of air supply and by managing the obstructions in the plenum. For example, Macedo and Alkharabaheh [11,12] noted that increasing the total air supply volume contributed to enhancing the uniformity of airflow distribution. In their research, they observed CPU temperatures changed up to 10 °C with increased air flowrate. Alissa et al. [13] suggested that obstructions should be placed in the “safe path” area, which can reduce the decay of airflow pressure by 16 % and lower temperature by 12.7 % compared to placing them in the “critical path” area. Airflow loss can be improved by reducing bypass loss and leakage loss. For instance, Zhou et al. [14] arranged rack baffles and server terminal baffles to decrease bypass loss, resulting in a reduction of hotspot temperature by 0.7 °C. Makwana et al. [15] placed sealing grommets under the racks, which led to a 15.2 % increase in supply airflow.

Enhancing temperature uniformity among high heat flux components is also crucial in single-phase immersion cooling systems. Cheng et al. [16] found that a faster coolant flow speed resulted in a more uniform temperature distribution around the heat sink. For the practical system, several servers are placed in a liquid cooling tank to dissipate heat [17]. However, current research in this area mainly concerns the coolants and performance optimization. Table 1 shows the current literature reviews about the single-phase immersion cooling system. For the coolant, the impact of thermophysical properties and the nanofluids are discussed. For example, Hnayno et al. [18] found that an increase in coolant viscosity from 4.6 to 9.8 mPa s led to a decrease in cooling performance of about 6 %. Rafati et al. [19] found that alumina nanofluid displayed superior thermal performance, with temperatures 5.5 °C lower than those

Table 2

Detailed parameters for the server.

(3) Design of liquid cooling tank

Item	Type	Description	Number
Server		Size: $340 \times 177 \times 683$ mm Maximum heat load	1 2800 W
GPU	RTX 4090 24 GB	Size: $26.5 \times 24 \times 0.6$ mm Maximum heat load	8 350W
CPU	Intel Platinum 8352V	Size: $40 \times 35 \times 9$ mm	2
GPU heat sink	Base plate	Size: $200 \times 122 \times 5$ mm	8
	Fins	Size: $200 \times 28 \times 0.4$ mm	107
Memory bank	Samsung DDR4 ECC	Size: $140 \times 30 \times 2$ mm	16
System drive	Gen3 Samsung PM983	Size: $100.2 \times 69.85 \times 6.8$ mm	1
Digital hard drive	Gen 4 Samsung PM9A3	Size: $22 \times 110 \times 4$ mm	3
Power supply	CRPS 2400W	Size: $185 \times 73.5 \times 39$ mm	4

achieved using a pure base fluid under identical operating conditions. For the performance optimization, different structures are discussed and evaluated. For example, Muneeshwaran et al. [20] found the thermal resistance and temperature of the T configuration were reduced by 12.6 % and 0.5–2.8 °C, respectively, compared to the Z configuration. Li et al. [21] discussed the use of heat sinks in single-phase immersion cooling systems. They suggested optimal structures, including fin number, fin thickness, fin height, and substrate thickness, which led to a 19 % reduction in heat sink mass and only a 4.5 % increase in temperature. Kuncoro et al. [22] used the Taguchi method to identify key parameters affecting immersion cooling performance. Their results indicated that cooling fans impacted CPU temperature by 71.3 %, and energy consumption by 75.0 %. Even though some works are conducted based on server racks in the tank, as shown in Table 1, the temperature uniformity among the CPUs or GPUs are ignored.

In the open literatures, the temperature difference among CPUs or GPUs could vary significantly, from as little as 1.5 °C to as much as 17 °C under different operating conditions [30,34]. For the practical application, the temperature uniformity should be clarified with the fact that there have more CPUs or GPUs with high power density in the liquid cooling tank, and a bad temperature uniformity would lead to the shutdown or even burnout of server; in addition, the flow distribution of coolant in the tank would largely affect the temperature uniformity. To bridge this knowledge gap, the performance of liquid cooling tank is designed and experimentally discussed, where 16 high power density servers are arranged (700–2800 W per server). The impacts of inlet flowrate, inlet temperature, and load ratio of server on the average GPU temperature, outlet temperature, and deviation and square deviation of GPU temperatures are explored. The conclusion obtained in this study could provide the guidance for the temperature uniformity characteristics and designing the liquid cooling tank.

2. Design of the liquid cooling tank and system description

2.1. Design of the liquid cooling tank

To design the liquid cooling tank, the performance of single server is firstly conducted through experiment to obtain the temperature variation and pressure variation along with the coolant flowrate. Then the simplified CFD model of single server is established and validated. Finally, the structure of the tank, as well as the system, is designed and manufactured.

(1) Experimental study for the single server

The server with a high-power density, equipped with eight GPUs (Model: RTX 4090 24 GB) and two CPUs (Model: Intel Platinum 8352V), is used, as depicted in Fig. 1(a). The total power for one server is as high as 2800 W. To enhance the distribution of coolant flow, baffles have been integrated into the design. Additionally, the pin fin heat sinks on the CPU and GPU are used to facilitate more effective heat dissipation [21]. The detailed specifications of the server's components are listed in Table 2. The experiment is firstly conducted to explore the operational characteristics for such server. During the experiment, the inlet temperature and the load ratio are set as 25 °C and 100 %, respectively. And the flowrate of coolant ranges from 0.13 to 2.19 m³/h. The average temperature for eight GPUs and the pressure loss within the server are monitored and employed to evaluate the server performance.

(2) Server simplification

Since there have many components in server, it is hard to regard all the components when design the liquid cooling tank. In this study, 16 servers would be arranged in the liquid cooling tank. Therefore, the server simplification is necessary. Based on the experimental data, the relationship between the flowrate and the average temperature for eight GPUs and pressure loss within the server are fitted as empirical equation, as shown in Eqn (1). Then, the server is simplified as the equivalent model in Computational Fluid Dynamics (CFD) simulations, as shown in Fig. 1(b). Such model allows to simulate temperature variations and pressure losses within the system under different flowrates. The simulation results have been validated with the experimental data and are illustrated in Fig. 1(c). This validated model enables the streamlined design of liquid cooling tanks by providing a more manageable

Table 3
Detailed parameters for the liquid cooling tank.

Item	Type	Size (mm)	Number
Tank	Length $\times$ width $\times$ height	1494 $\times$ 926 $\times$ 850	1
Porous panel	Length $\times$ width $\times$ height	1488 $\times$ 802 $\times$ 2	
	Holes	R15	128
Delivery pipe	Manifold	R26 $\times$ 1344	2
	Branch pipes	R16 $\times$ 708	2
	Inlet	R54	1
	Outlets	R12	216

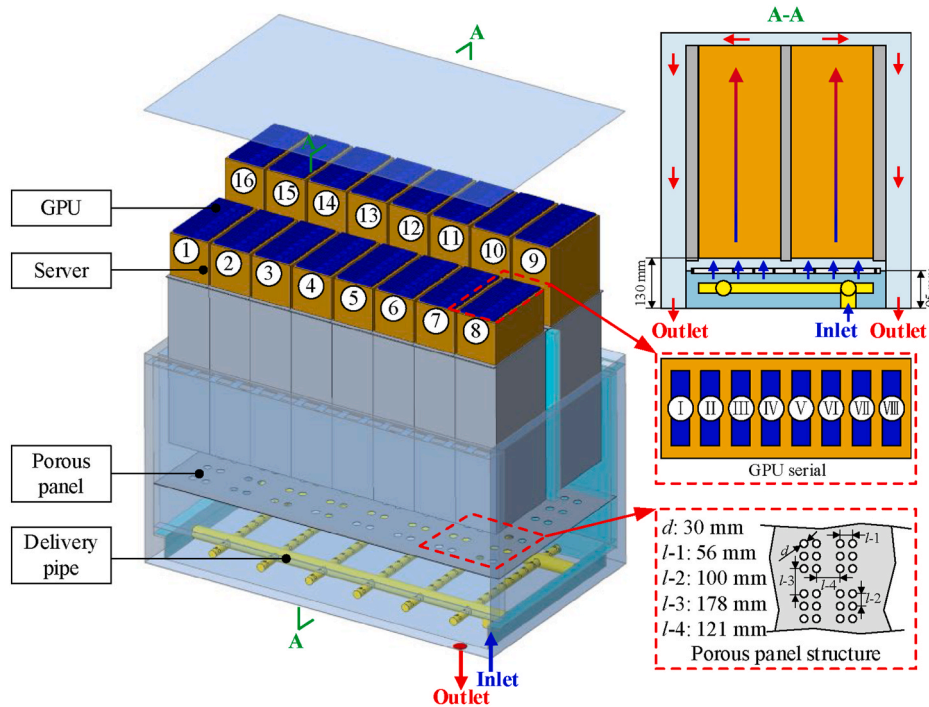


Fig. 2. Schematic of liquid cooling tank.

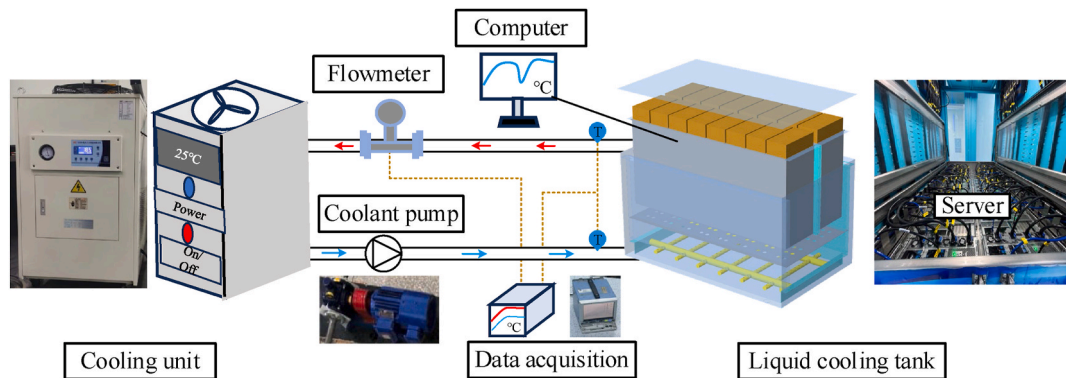


Fig. 3. Schematic of the experimental system.

Table 4

Detailed parameter for the experimental system.

Item	Type	Range	Uncertainty
Thermocouples	TT-K-30	−60–300 °C	±0.1 °C
Temperature for GPUs [36]	Digital sensor	−40–125 °C	±1 °C
Flowmeter	KL-LUXB	0–46 m ³ /h	±0.01m ³ /h
Cooling unit	COF-40AP	−30–100 °C	±0.1 °C
Coolant pump	TYPE40-30	5–32m ³ /h	
Data acquisition	Yokogawa-GP10	/	/

Table 5

Thermophysical parameter of coolant.

Temperature (°C)	Density (kg/m ³)	Heat capacity (J/(kg·°C))	Dynamic viscosity (Pa·s)	Thermal conductivity (W/(m·°C))
20	794	1965	0.0117	0.34
40	786	2051	0.0061	0.36
60	775	2154	0.0036	0.38
80	762	2253	0.0024	0.42

representation of the server's thermal behavior.

$$\begin{aligned} T &= 0.0697q^2 - 1.3246q + 61.257 \\ \Delta P &= 0.03q^2 + 0.538q + 0.14 \end{aligned} \quad (1)$$

where q is the coolant flowrate, m³/h; T is the average temperature of eight GPUs; ΔP is the pressure loss within the server.

The liquid cooling tank's primary function is to ensure that the coolant is distributed evenly across all servers within the cooling system. In this study, 16 servers are organized into two rows in the liquid cooling tank. By using the simplified server model, the impact of different parameters on the flow distribution in the liquid cooling tank is conducted, including the manifold, the branch pipe, the porous panel, the separation chamber, and the flowrate on the flow distribution, ensuring that the coolant effectively reaches all parts of each server. More detailed analysis process can be found in our previous study [35]. Based on the analysis, the detailed parameters for the tank can be found in Table 3. In addition, the designed tank and the serials for server and GPUs can be found in Fig. 2. Since there have 128 GPUs in the liquid cooling tank, the variation of GPU temperatures is employed to assess the performance of liquid cooling tank.

2.2. System description

Based on the designed liquid cooling tank, the system is also established, which comprises of the liquid cooling tank, 16 high power density servers, the coolant pump, the flowmeter, and the cooling unit, as shown in Fig. 3. During the operation, the pump drives the coolant to the liquid cooling tank where the coolant would be evenly transferred to each server through the delivery pipe and porous panel. Then the heat produced in server could be removed timely by the circulation of coolant. Thus, the heat in coolant with high temperature could be transferred to the ambient in coolant cooling unit. The detailed parameter for each component, as well as its uncertainty, can be found in Table 4. The thermocouples are arranged at the inlet and outlet pipes to obtain the coolant temperature. For the GPU temperatures, they can be obtained in computer since the temperature sensor are embedded on GPUs. During the experiment, the impact of the coolant flowrate, the inlet temperature of the coolant, and the load ratio of the server on the average GPU temperature is conducted.

The coolant used in this study is “BINGYI 797”, which is produced by Tianjin Tier Technology Co. Ltd. The thermophysical properties of the coolant used in this study can be found in Table 5.

2.3. Key performance indicators

To assess the performance of liquid cooling tank in the experiment, the average GPU temperature, the outlet temperature of the liquid cooling tank, and the deviation and the standard deviation of the GPU temperatures are used.

The average GPU temperature ($\bar{T}$) is the mean temperature of 128 GPUs in the experiment, which could be directly obtained in the computer.

The outlet temperature could be obtained through the data acquisition by using thermocouples.

The temperature deviation is the temperature difference between the maximum temperature and minimum. It indicates the spread in temperature across different GPUs in the system, which can be calculated as follows:

$$T_d = \max(T_i | i = 1, 2, 3 \dots, 128) - \min(T_i | i = 1, 2, 3 \dots, 128) \quad (2)$$

The square deviation of the GPU temperature could assess the temperature uniformity in the liquid cooling tank. It is a measure of how much the GPU temperatures vary from the average value. The relatively low deviation and square deviation suggest that the

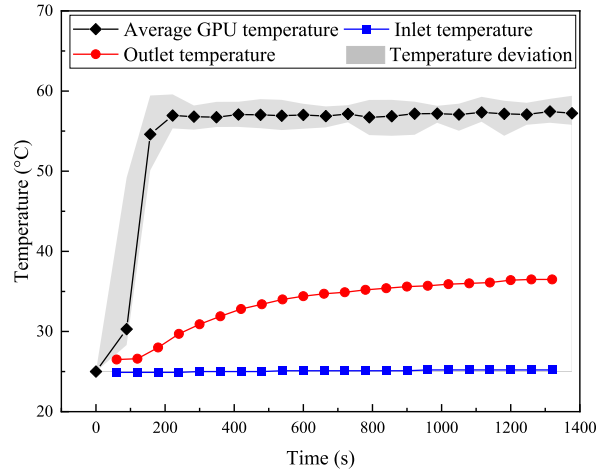


Fig. 4. Temperature variation during the operation (Inlet temperature = 25 °C, flowrate = 10 m³/h, load ratio = 100 %).

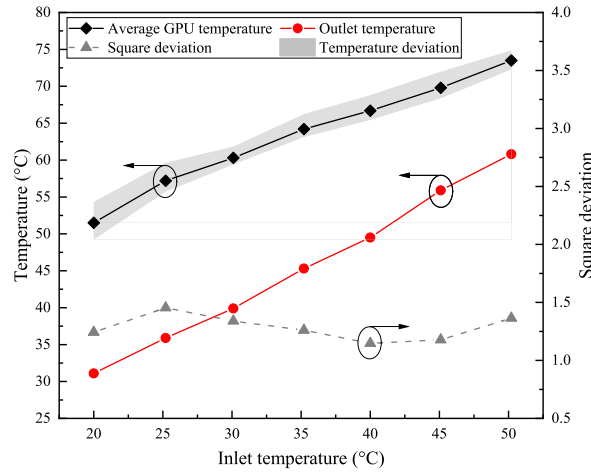


Fig. 5. Impact of inlet temperature on the performance of liquid cooling tank (Flowrate = 10 m³/h, load ratio = 100 %).

cooling system maintains uniform temperature across all GPUs, which is desirable for consistent performance and reliability. The square deviation can be calculated as follows:

$$\sigma_{S,D} = \frac{1}{N} \sum_{i=1}^N (T_i - \bar{T})^2 \quad (n = 128, i = 1, 2, \dots, 128) \quad (3)$$

where N is the number of GPU, T_i is the temperature of GPU i , $\bar{T}$ is the mean of all temperature of GPU

3. Results and discussion

3.1. Result for the system operation

Fig. 4 illustrates the temperature variation of the liquid cooling system during operation. The average GPU temperature rapidly increased at the beginning of the system's operation, indicating the heat was produced on GPUs. Meanwhile, the outlet temperature changed slightly within 150 s which was due to the coolant circulation. After this initial increase, the average GPU temperature stabilized at around 57 °C after 200 s. This plateau indicated that the heat exchange by the coolant became balanced with the heat generation by the GPUs, reaching a steady-state condition. However, the outlet temperature was increased more slowly compared to the GPU temperature and stabilized at 36 °C at around 1000 s. This slower increase was likely due to the thermal mass of the coolant and the system's ability to continuously remove heat. The temperature difference between the inlet and outlet (around 11 °C) demonstrated the amount of heat absorbed by the coolant as it passed through the system. For the temperature uniformity, the maximum temperature deviation lied in the start-up process (within 200 s), which was due to the different heat loads of GPUs during

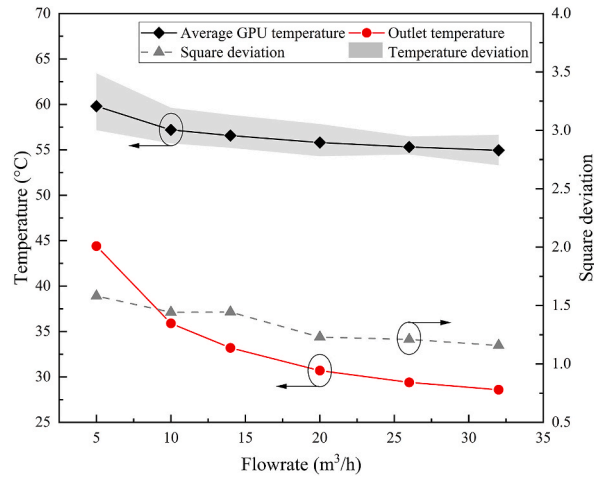


Fig. 6. Impact of coolant flowrate on the performance of liquid cooling tank (Inlet temperature = 25 $^{\circ}\text{C}$, load ratio = 100 %).

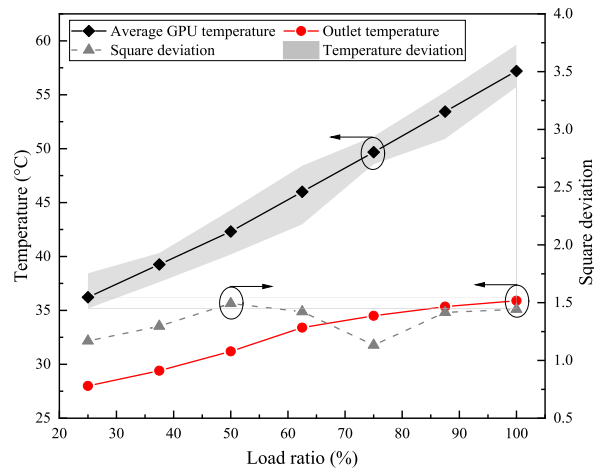


Fig. 7. Impact of load ratio on the performance of liquid cooling tank (Inlet temperature = 25 $^{\circ}\text{C}$, flowrate = 10 m^3/h).

this stage. If all the GPUs reached the maximum heat load, the temperature deviation was within 4.8 $^{\circ}\text{C}$, with a square deviation of 1.44, indicating such liquid cooling tank showed a good temperature uniformity.

3.2. Impact of different parameters on tank performance

Fig. 5 shows the impact of inlet temperature on the liquid cooling tank. Apparently, the outlet temperature linearly increased with the increase of inlet temperature, but their temperature difference was maintained around 11 $^{\circ}\text{C}$, indicating the heat could also be removed timely. However, the temperature variation for average GPU temperature was different. When the inlet temperature increased from 20 $^{\circ}\text{C}$ to 50 $^{\circ}\text{C}$, the outlet temperature was increased by 30 $^{\circ}\text{C}$; however, the average GPU temperature was only increased by 21 $^{\circ}\text{C}$. This phenomenon was mainly due to the variation of thermophysical of the coolant. Through Tables 5 and it can be found the dynamic viscosity decreased by 79.5 % and thermal conductivity increased by 23.5 % when the temperature increased from 20 $^{\circ}\text{C}$ to 80 $^{\circ}\text{C}$, which was benefit to the coolant flow and heat exchange. Moreover, it can be found the average GPU temperature was 73.5 $^{\circ}\text{C}$ even when the inlet temperature reached as high as 50 $^{\circ}\text{C}$, indicating such cooling system could take full advantage of free cooling and the energy consumption should be largely saved. For the temperature uniformity, it can be found the temperature deviation was within 4.8 $^{\circ}\text{C}$ and the square deviation was around 1.0–1.5, indicating such cooling tank showed a good flow distribution and the inlet temperature had slight impact on the temperature uniformity.

Fig. 6 shows the impact of coolant flowrate on the performance of liquid cooling tank. With the increase of coolant flowrate, the heat produced in server can be removed timely, and therefore, both the average GPU temperature and the outlet temperature decreased. It should be note that, when the flowrate increased from 5 m^3/h to 10 m^3/h , the average GPU temperature sharply decreased from 59.8 $^{\circ}\text{C}$ to 54.9 $^{\circ}\text{C}$, by 4.9 $^{\circ}\text{C}$, indicating the increase of flowrate was an effective way to decrease the GPU temperature.

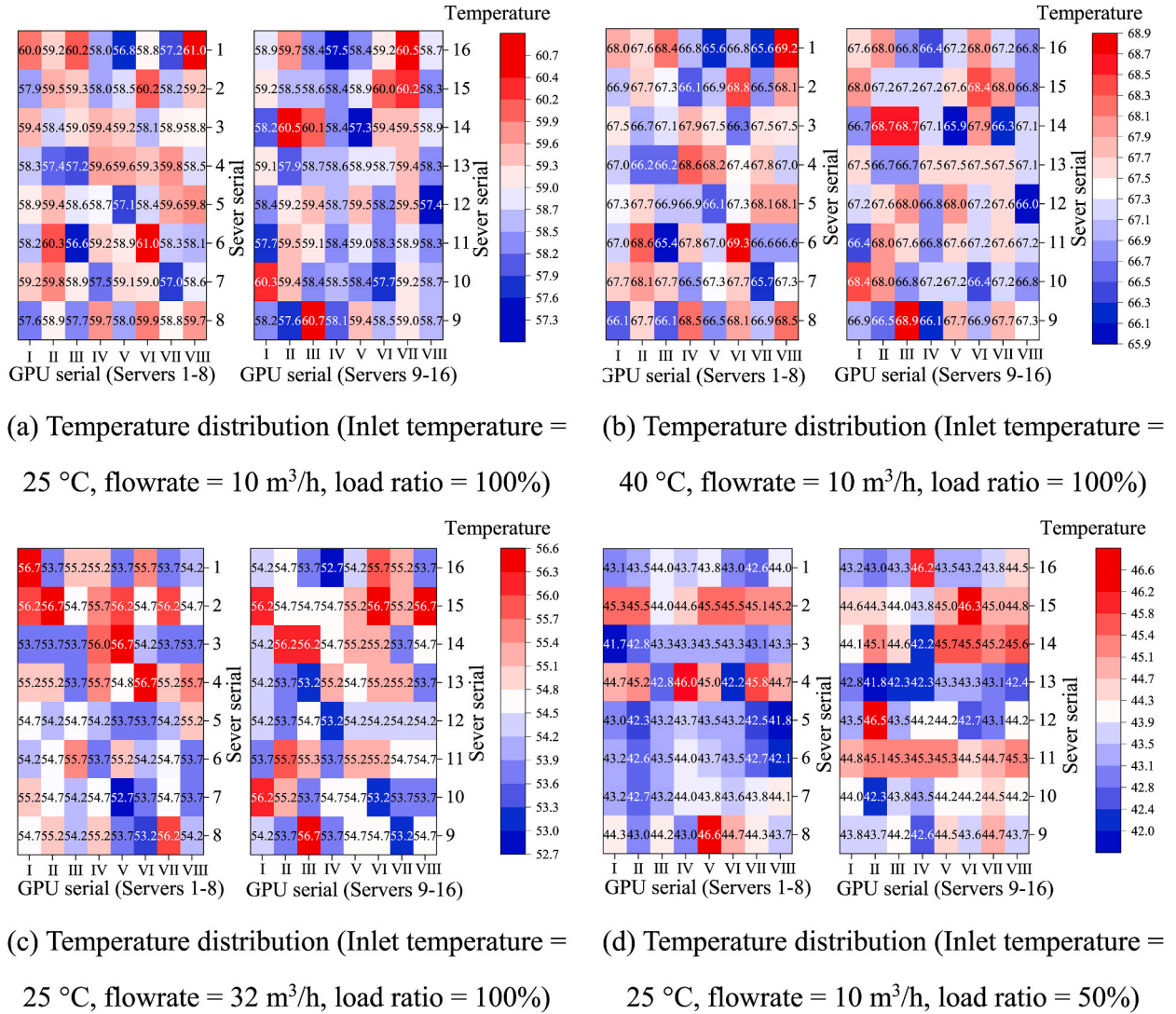


Fig. 8. Temperature distribution in the liquid cooling tank for different conditions.

However, if the flowrate was large enough, the reduction of GPU temperature was not obvious. In this study, the average GPU temperature was only decreased by 2.2 °C when the flowrate increased from 10 m³/h to 32 m³/h. Similar results can also be found in Luo's work [37]. Moreover, the large coolant flowrate indicated more energy would be consumed. In addition, a large flowrate was benefit to enhance the temperature uniformity. When the flowrate was 5 m³/h, the temperature deviation can be as high as 6.3 °C, with a square deviation of 1.58; while, these values were decreased to 4.1 °C and 1.15, respectively when the flowrate increased to 32 m³/h. Therefore, from the aspect of average GPU temperature, 10–15 m³/h of flowrate is recommended for such system.

Fig. 7 shows the impact of load ratio of GPUs on the performance of liquid cooling tank. Apparently, both the average GPU temperature and outlet temperature showed linearly increasing tendency with the increase of load ratio of server. The variation of average GPU temperature was more obvious, which was due to the heat produced was significantly different with different load ratio. For example, the average GPU temperature increased by 21.0 °C when the load ratio increased from 25 % to 100 %; while, this value was only 7.9 °C, which was mainly due to the unchanged flowrate of coolant. For temperature uniformity, the temperature deviation and the square deviation were within 5 °C and 1.5, respectively, indicating the load ratio had slight impact on the temperature uniformity.

3.3. Characteristics of the temperature distribution

Fig. 8 shows the temperature distribution about the 128 GPUs. Overall, the temperature deviation for each condition were all within 5 °C, indicating such liquid cooling tank showed a good performance on flow distribution. However, the “hot spot” can also be found on some GPUs, which could be divided into two categories. One is the higher temperature can be occurred for all different

operational conditions, such as Server-2-VI, Server-15-VI, etc. The reason can be attributed to the insufficient flowrate here and irrationality design of the tank. Another kind of “hot spot” occurred in some cases, such as Server-7-VI, Server-14-II, Server-9-III, etc. These may be due to the overclocking operation of GPU and the low accuracy of the digital temperature sensor from the GPU cores. These two different hot spots should be carefully considered when designing the liquid cooling tank.

3.4. Discussion

This study experimentally discussed the performance of liquid cooling tank used in data center, especially for the temperature uniformity of the GPUs. However, there have some works should be considered in the future: (1) the energy consumption analysis and optimization: in the system, the coolant pump, the water pump, and the fans for cooling device would be used. To maintain the safety operation of server and obtain the lowest PUE, the energy consumption would be conducted and optimized regarding the different zones and different control strategies. (2) Since the digital temperature sensor at GPU cores are used to obtain the temperature, more accurate temperature measurement methods would be explored based on ensuring the safe operation of server and data center.

4. Conclusions

The performance of the liquid cooling tank used in single-phase immersion cooling data center is experimental discussed. The impacts of inlet temperature, coolant flowrate, and the load ratios of server on the performance of liquid cooling tank are discussed, during which the average GPU temperature, the outlet temperature, the deviation and square deviation for GPU temperatures are employed as key performance indicators. Through the results, it can be concluded as follows:

- (1) Decreasing inlet temperature of coolant and the load ratio, and increasing the flowrate are benefit to decrease the average GPU temperature.
- (2) The decrease of dynamic viscosity and the increase of thermal conductivity is also helpful to the heat exchange. When the inlet temperature increased from 20 °C to 50 °C, the outlet temperature was increased by 30 °C; however, the average GPU temperature was only increased by 21 °C.
- (3) The designed liquid cooling tank showed a good temperature uniformity, in which the temperature deviation was within 5 °C and the square deviation was in the range of 1.0–1.5.

CRediT authorship contribution statement

Xueqiang Li: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Shentong Guo:** Writing – review & editing, Data curation, Investigation. **Haiwang Sun:** Writing – review & editing, Investigation, Conceptualization. **Shen Yin:** Writing – review & editing. **Shengchun Liu:** Methodology, Conceptualization, Supervision. **Zhiming Wang:** Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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